Brief explanation of the contents of the 2nd Deep Data Releases

Deep Data Release 2 consists of the full-depth data of the following 9 MHONGOOSE galaxies: J0135-41, J0335-24, J0429-27, J0459-26, J0546-52, J1103-23, J2009-61, J2257-41, J2357-32.

Note that J2357-23 contains Magellanic Stream HI overlapping with the galaxy. This release therefore includes moment maps with the Stream HI either included (`J2357-32_orig`) or removed (`J2357-32`).

When using these data please include a reference to the MHONGOOSE survey paper: de Blok et al (2024), A&A, **688**, A109; https://ui.adsabs.harvard.edu/abs/2024A%26A...688A.109D/abstract

Please see the webpage at https://mhongoose.astron.nl/sample.html and the MHONGOOSE survey paper for further details on the global properties of these galaxies.

The release contains three main directories, containing the data cubes, the moment maps (including primary beam maps and masks), and data release documents (including Atlas pages and Legacy Survey overlays). It is recommended to first have a look at the PDF files in the `documents` folder to familiarise yourself with the galaxies prior to downloading large numbers of data sets. A brief description of the contents of each directory is given below.

Cubes

We provide data cubes for the target galaxies at six standard resolutions. The cubes have a velocity resolution of 1.4 km/s and a velocity range of ~1000 km/s straddling the velocity of the target galaxy. The actual range may vary slightly depending on whether the cube contains any flagged Milky Way channels. The cubes measure 1.5 x 1.5 degrees in angular size. This is larger than the MeerKAT primary beam of ~1 degree due to the significant number of sources that can be detected in the outer field for some of the galaxies.

In addition to the full data reduction as described in the survey paper, we added an extra step, consisting of a final residual continuum subtraction in the image plane using the `imcontsub` routine developed by Sphesihle Makhathini (`https://github.com/laduma-dev/contsub`). This is done *after* the creation of the moment maps as described in the survey paper. In this step, we first use `imcontsub` to mask the data cubes using the 3D moment map masks created by `SoFiA-2` (thus removing all detected emission). A low-order polynomial is then fitted to all spectra in the masked cubes. These fitted cubes are subtracted from the respective image cubes, after which new moment maps are created using `SoFiA-2` (using the same parameter file). The difference between initial and final moment maps is negligible, but the procedure does improve the statistics of the noise, and automatically removes residual continuum features that are not visible in individual channels but were detected by `SoFiA-2` in the initial moment-map run.

We include cubes at various resolutions, covering the HI resolution range available to MeerKAT, created using a combination of robust weightings and taperings. See the MHONGOOSE survey paper and specifically Table 4 therein, which is reproduced below. The actual beam sizes can be found in the headers, and the noise can be directly measured in the relevant channels. Small variations with respect to the indicative values should be expected, depending on declination, effective observing time, amount of data flagged etc. Due to their large size the `r00_t00` cubes are not included, but they are available upon request. All relevant features in the `r00_t00` cubes are also visible in the `r05_t00` cubes at slightly reduced resolution.

File names contain the galaxy HIPASS identification (`J*`), the robust value (`r??`) and the taper value (`t??`). These are also listed in the Table below.

de Blok, W. J. G., et al.: A&A, 688, A109 (2024)

Table 4. Standard resolutions.

Label (1)	Robust value (2)	Taper ($''$) (3)	Pixel ($''$) (4)	b_{maj} ($''$) (5)	b_{min} ($''$) (6)	b_{PA} ($^\circ$) (7)	Noise (mJy beam^{-1}) (8)	$\log N_{\mathrm{HI}}^{1\sigma,1\mathrm{ch}}$ (cm^{-2}) (9)	$\log N_{\mathrm{HI}}^{3\sigma,16\,\mathrm{km\,s^{-1}}}$ (cm^{-2}) (10)	$\log M_{\mathrm{HI}}$ ($M_\odot$) (11)
r10_t90	1.0	90	30	94.2	91.2	47	0.318 ± 0.011	16.76	17.77	6.05
r05_t60	0.5	60	20	65.3	64.0	92	0.250 ± 0.009	16.97	17.98	5.95
r15_t00	1.5	0	7	34.4	25.4	135	0.154 ± 0.004	17.44	18.44	5.74
r10_t00	1.0	0	5	26.4	18.2	136	0.150 ± 0.004	17.69	18.69	5.73
r05_t00	0.5	0	3	14.1	9.7	137	0.171 ± 0.005	18.29	19.29	5.78
r00_t00	0.0	0	2	8.2	7.1	142	0.219 ± 0.007	18.77	19.77	5.89

Notes. (1) Label used to refer to this resolution. (2) Robustness weighting value. (3) Taper used. "0" mean no taper. (4) Pixel size of the data cubes and moment maps. (5) Average major axis beam size. (6) Average minor axis beam size. (7) Average position angle of major axis of beam. (8) Noise per channel. (9) 1σ, one-channel column density sensitivity. (10) 3σ, $16\,\mathrm{km\,s^{-1}}$ column density sensitivity. (11) H I mass sensitivity for a 3σ, $50\,\mathrm{km\,s^{-1}}$ unresolved source at 10 Mpc.

Moment Maps

We provide zeroth, first and second moment maps at all standard resolutions for the target galaxies. These were produced using SoFiA-2 based on the `imcontsub` cubes (see the *Cubes* section above). The most important details are that we used the S+C finder with spatial kernels of 0, 4 pixels, spectral kernels of 0, 9, 25 channels, and a threshold of 4 times the local rms. In terms of reliability we used a `minSNR` value of 5.0 and a reliability threshold of 0.8. The latter value was usually not relevant, as in the very large majority of cases the detected objects were very clearly separated from the reliability noise distribution, so that the value of `minSNR` was the only relevant parameter. In terms of kernels we found that using larger spatial kernels than used here led to over-smoothing and the addition of noise in the moment maps (i.e., MeerKAT is very efficient at picking up HI emission and large smoothing factors give diminishing returns). More information can also be found in the Survey Paper.

Moment maps are organised as follows: for each galaxy there are six subdirectories corresponding to each of the standard resolutions. In each of these are the respective moment maps (`*mom0*`, `*mom1*`, `*mom2*`), a map showing the number of channels contributing to a moment-map pixel (`*chan*`), a two dimensional mask identifying the detected sources (`*mask-2d*`) and a (compressed) 3D mask that was used to create the moment maps (`*mask.fits.gz`; except for `r00_t00`). In addition a zeroth-moment map which has been primary-beam corrected is also provided (`*mom0_pb*`) as well as the primary beam map used (`*primbeam*`). We only provide a primary-beam image and not a primary-beam cube as any change in the size of primary beam is negligible over the velocity ranges presented here.

Atlas Pages

We provide Atlas pages in PDF format for a quick visual overview of the HI distribution in these galaxies. These can be found in the `documents/Atlas` directory

Top row, first panel: Optical colour image created from the Dark Energy Camera Legacy Survey (DECaLS) DR10 release.

Top row, second panel: Zeroth-moment (integrated intensity) map of the target galaxy based on `r00_t00` data (high resolution). The lowest contour level corresponds to the S/N=3 level (see below). Contours then increase by a factor 2^n where n=0,2,4,… . Values are given in the legend in the panel. The grey scale runs from 0 to 99.99 percentile with a `sqrt` stretch. The map displayed here is primary-beam corrected.

Top row, third panel: Zeroth-moment map of the target galaxy based on `r10_t00` data (medium resolution). The lowest contour level corresponds to the S/N=3 level (see below). Contours are then spaced as 2^n where n=0,2,4,… . Values are given in the legend in the panel. The grey scale runs from 0 to 99.99 percentile with a `sqrt` stretch. The map displayed here is primary-beam corrected.

Top row, fourth panel: First-moment (intensity-weighted velocity) map of the target galaxy based on `r10_t00` data. The thick black contour indicates an indicative systemic velocity as determined by the parametrisation by SoFiA-2. Contours are then spaced by 5, 10, or 20 km/s as indicated in the panel. Blue colours indicate the approaching side, red colours the receding side. The indicative systemic velocity used here does not necessarily represent the actual systemic velocity of the target galaxy.

Bottom-left panel: multi-resolution HI map of the target galaxy. This shows the S/N=3 contours (see below) for a range of resolutions, derived from different weighting and tapering combinations. The column density that belongs to each of the contours is indicated in the colour bar at the corresponding colour. The geometric mean size of the beam size is also given there. The background image is the *r*-band DR10 legacy survey image.

Centre-right panel: position-velocity slice along an indicative major axis position angle. The value of the PA is indicated in the panel. This value does not necessarily equal the proper kinematical PA as would be derived from a full kinematical analysis. The grayscale contours have values 3*`rms`*2^n, where n=0,1,2,…. The blue contour indicates 3*`rms`, the red contour -2.5*`rms`. The horizontal green line show the indicative systemic velocity. The grayscale runs from -1.5*rms to the 99.95 percentile with an `asinh` stretch.

Bottom-right panel: Low-resolution integrated HI map (based on `r05_t60`) encompassing all sources detected in the field. The lowest (blue) contour is the S/N=3 contour. Levels then increase by factors 2^n, where n=0,2,4,… .

The S/N=3 contour
In the zeroth moment panels, the S/N=3 contour level is specified in the text at the bottom of the panel. The S/N=3 contour is calculated by using an S/N map which is defined as the ratio of the zeroth-moment map and a noise map. The latter is calculated as `[rms * dv * sqrt(nchan)]`, where `dv` is the channel width of the cube, `nchan` is

the number of channels in that particular line of sight that contribute to the zeroth moment map and `rms` is the noise in a single channel. The column density associated with S/N=3 is then calculated by taking the median column density of the pixels which have S/N values in the S/N map between 2.75 < S/N < 3.25.

Multi-resolution overlays

For each galaxy the `documents/MultiRes` directory contains multi-resolution HI overlays on optical *grz* Legacy Survey images. The contour levels for each resolution are listed in the bottom-left of the figure and are again multiples of the S/N=3 contour as described above for the Atlas pages. Yellow: `r10_t90`; purple: `r05_t60`; blue: `r15_t00`; green: `r10_t00`; red: `r05_t00`.

The HI colour image overlay in yellow-orange is created by combining the zeroth-moment maps of the above standard resolutions using an algorithm that mimics the "screen" layer overlay option in Gimp image editing software. The HI overlay image was created to simultaneously show the low- and high-column density features and the image values should not be used for quantitive purposes.

For each galaxy a number of different versions of its multi-resolution image are included:

`*_multihi_decals_r05` = this shows an overlay of the combined HI column density image on the grz optical image. The lower column densities are represented by contours from each of the standard resolutions as described above.

`*_multihi_decals_r05` = as the previous image, except the `r05_t00` contour is not plotted to more clearly show the HI image.

`*_multihi_decals_nocontours` = only the HI image overlay is shown and no contours.

`*_nomultihi_decals_nor05` = neither HI image, nor the `r05_t00` contours are plotted to better show the optical galaxy.